PHYDSC402T — Classical Mechanics

Unit 5: Rigid Body Dynamics and Small Oscillations

7th Semester, Assam University (FYUG) — Haflong Government College

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Notes 11 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Motion of rigid bodies: Angular momentum and kinetic energy, inertia tensor, principal axes and moments of inertia.
2. Euler's angles, Euler's equations of motion.
3. Coriolis force.
4. Force-free motion of a symmetrical top.
5. Small oscillations: equilibrium and potential energy, frequencies of free vibration and normal coordinates.
6. Longitudinal vibration of linear triatomic molecule.

Top 2 high-mark topics in this unit

1. Euler angles and Euler's equations of motion for a rigid body, including force-free motion of a symmetrical top — the main rigid-body derivation, repeated in Christ College, St. Aloysius and CBCS-pattern papers.
2. Small oscillations: equilibrium and potential energy, frequencies of free vibration, normal coordinates, and longitudinal vibration of a linear triatomic molecule — the second half of the unit, derivation plus normal modes, repeated as triatomic-molecule and coupled-pendulum questions.

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1. Angular Momentum and Kinetic Energy of a Rigid Body

A rigid body is defined as a system of particles in which the distance between any pair of constituent particles remains fixed over time. Consider a rigid body composed of discrete mass points m_α positioned at vectors $\mathbf{r}_\alpha$ with respect to a chosen origin O fixed in space or at the body's center of mass.

The total angular momentum $\mathbf{L}$ of the rigid body about the origin O is defined as the vector sum of the angular momenta of all individual particles:

$$\mathbf{L} = \sum_\alpha \mathbf{r}_\alpha \times \mathbf{p}_\alpha$$

Substituting the linear momentum of the α-th particle, $\mathbf{p}_\alpha = m_\alpha \mathbf{v}_\alpha$, into the angular momentum equation yields:

$$\mathbf{L} = \sum_\alpha m_\alpha (\mathbf{r}_\alpha \times \mathbf{v}_\alpha)$$

When the rigid body rotates with instantaneous angular velocity $\boldsymbol{\omega}$ about a fixed point at the origin O, the velocity $\mathbf{v}_\alpha$ of the α-th particle relative to the origin is given by:

$$\mathbf{v}_\alpha = \boldsymbol{\omega} \times \mathbf{r}_\alpha$$

The total kinetic energy T of the rigid body is the sum of the kinetic energies of all its individual particles:

$$T = \frac{1}{2} \sum_\alpha m_\alpha v_\alpha^2$$

Expressing the square of the velocity as a scalar dot product $\mathbf{v}_\alpha \cdot \mathbf{v}_\alpha$, the total kinetic energy takes the form:

$$T = \frac{1}{2} \sum_\alpha m_\alpha (\mathbf{v}_\alpha \cdot \mathbf{v}_\alpha)$$

Substituting $\mathbf{v}_\alpha = \boldsymbol{\omega} \times \mathbf{r}_\alpha$ into one factor of the dot product gives:

$$T = \frac{1}{2} \sum_\alpha m_\alpha \mathbf{v}_\alpha \cdot (\boldsymbol{\omega} \times \mathbf{r}_\alpha)$$

Using the vector triple product identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A})$, the term inside the summation can be rearranged as:

$$\mathbf{v}_\alpha \cdot (\boldsymbol{\omega} \times \mathbf{r}_\alpha) = \boldsymbol{\omega} \cdot (\mathbf{r}_\alpha \times \mathbf{v}_\alpha)$$

Substituting this identity back into the kinetic energy expression yields:

$$T = \frac{1}{2} \sum_\alpha m_\alpha \boldsymbol{\omega} \cdot (\mathbf{r}_\alpha \times \mathbf{v}_\alpha)$$

Since the angular velocity $\boldsymbol{\omega}$ is common to all particles within the rigid body, it can be factored out of the summation:

$$T = \frac{1}{2} \boldsymbol{\omega} \cdot \sum_\alpha (\mathbf{r}_\alpha \times m_\alpha \mathbf{v}_\alpha)$$

Recognizing the summation term as the total angular momentum $\mathbf{L} = \sum_\alpha \mathbf{r}_\alpha \times m_\alpha \mathbf{v}_\alpha$, the total kinetic energy of a body rotating about a fixed point simplifies to:

$$T = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{L}$$

2. Inertia Tensor, Principal Axes and Moments of Inertia

Inertia Tensor Definition and Components

To express the angular momentum and kinetic energy in terms of the components of the angular velocity, we substitute the relative velocity expression $\mathbf{v}_\alpha = \boldsymbol{\omega} \times \mathbf{r}_\alpha$ into the total angular momentum expression:

$$\mathbf{L} = \sum_\alpha m_\alpha [\mathbf{r}_\alpha \times (\boldsymbol{\omega} \times \mathbf{r}_\alpha)]$$

Applying the vector triple product identity $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$ to the expression inside the brackets gives:

$$\mathbf{r}_\alpha \times (\boldsymbol{\omega} \times \mathbf{r}_\alpha) = \boldsymbol{\omega}(\mathbf{r}_\alpha \cdot \mathbf{r}_\alpha) - \mathbf{r}_\alpha(\mathbf{r}_\alpha \cdot \boldsymbol{\omega})$$

Using Cartesian coordinate notation where $\mathbf{r}_\alpha = (x_{\alpha,1}, x_{\alpha,2}, x_{\alpha,3})$ and $\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)$, the i-th component of this vector product is written as:

$$[\mathbf{r}_\alpha \times (\boldsymbol{\omega} \times \mathbf{r}_\alpha)]_i = \omega_i r_\alpha^2 - x_{\alpha,i} \sum_{j=1}^{3} x_{\alpha,j} \omega_j$$

Using the Kronecker delta symbol δ_{ij}, which satisfies $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$, the angular velocity component ω_i can be written as $\omega_i = \sum_{j=1}^{3} \delta_{ij} \omega_j$:

$$[\mathbf{r}_\alpha \times (\boldsymbol{\omega} \times \mathbf{r}_\alpha)]_i = \sum_{j=1}^{3} \left(\delta_{ij} r_\alpha^2 - x_{\alpha,i} x_{\alpha,j} \right) \omega_j$$

Substituting this into the i-th component of the total angular momentum yields:

$$L_i = \sum_\alpha m_\alpha \sum_{j=1}^{3} \left(\delta_{ij} r_\alpha^2 - x_{\alpha,i} x_{\alpha,j} \right) \omega_j$$

Interchanging the order of summation gives:

$$L_i = \sum_{j=1}^{3} \left[\sum_\alpha m_\alpha \left(\delta_{ij} r_\alpha^2 - x_{\alpha,i} x_{\alpha,j} \right) \right] \omega_j$$

The quantity inside the square brackets defines the elements I_{ij} of the inertia tensor:

$$I_{ij} = \sum_\alpha m_\alpha \left(\delta_{ij} r_\alpha^2 - x_{\alpha,i} x_{\alpha,j} \right)$$

With this definition, the i-th component of angular momentum is expressed in compact form as:

$$L_i = \sum_{j=1}^{3} I_{ij} \omega_j$$

The inertia tensor $\mathbf{I}$ is represented as a symmetric 3×3 matrix, since $I_{ij} = I_{ji}$. Explicitly, the diagonal components (moments of inertia) are given by:

$$I_{xx} = \sum_\alpha m_\alpha (y_\alpha^2 + z_\alpha^2)$$

$$I_{yy} = \sum_{\alpha} m_{\alpha}(x_{\alpha}^2 + z_{\alpha}^2)$$

$$I_{zz} = \sum_{\alpha} m_{\alpha}(x_{\alpha}^2 + y_{\alpha}^2)$$

The off-diagonal components (products of inertia) are given by:

$$I_{xy} = I_{yx} = -\sum_{\alpha} m_{\alpha} x_{\alpha} y_{\alpha}$$

$$I_{xz} = I_{zx} = -\sum_{\alpha} m_{\alpha} x_{\alpha} z_{\alpha}$$

$$I_{yz} = I_{zy} = -\sum_{\alpha} m_{\alpha} y_{\alpha} z_{\alpha}$$

Substituting the component form of angular momentum $L_i = \sum_{j=1}^{3} I_{ij}\omega_j$ into the kinetic energy relation $T = \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{L} = \frac{1}{2}\sum_{i=1}^{3} \omega_i L_i$ yields the quadratic form for rotational kinetic energy:

$$T = \frac{1}{2}\sum_{i=1}^{3}\sum_{j=1}^{3} I_{ij}\omega_i\omega_j$$

Principal Axes and Moments of Inertia

Because the inertia tensor I_{ij} is a real symmetric 3×3 matrix, linear algebra guarantees that it can be diagonalized by choosing a suitable orthogonal set of coordinate axes fixed in the body, known as the principal axes of inertia.

In this principal coordinate system, all products of inertia vanish ($I_{ij} = 0$ for $i \neq j$), and the matrix takes a diagonal form. The corresponding diagonal elements are called the principal moments of inertia, denoted by I_1, I_2, and I_3.

In the principal axis coordinate frame, the components of angular momentum decouple into direct linear relations with the corresponding angular velocity components:

$$L_1 = I_1\omega_1$$

$$L_2 = I_2\omega_2$$

$$L_3 = I_3\omega_3$$

Similarly, the expression for the total kinetic energy reduces to a sum of decoupled quadratic terms:

$$T = \frac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right)$$

Worked Example: Inertia Tensor and Principal Axes for Four Mass Points

Consider a system consisting of four mass points, each of mass m, located at the Cartesian positions $(a, 0, 0)$, $(0, a, 0)$, $(0, 0, a)$, and (a, a, a) relative to the origin.

To evaluate the components of the inertia tensor about the origin, we compute the squared radial distances $r_{\alpha}^2 = x_{\alpha}^2 + y_{\alpha}^2 + z_{\alpha}^2$ for each particle $\alpha = 1, 2, 3, 4$:

$$r_1^2 = a^2 + 0^2 + 0^2 = a^2$$

$$r_2^2 = 0^2 + a^2 + 0^2 = a^2$$

$$r_3^2 = 0^2 + 0^2 + a^2 = a^2$$

$$r_4^2 = a^2 + a^2 + a^2 = 3a^2$$

Calculating the diagonal element $I_{xx} = \sum_\alpha m(y_\alpha^2 + z_\alpha^2)$ gives:

$$I_{xx} = m(0^2 + 0^2) + m(a^2 + 0^2) + m(0^2 + a^2) + m(a^2 + a^2)$$

Simplifying each term inside the summation:

$$I_{xx} = 0 + ma^2 + ma^2 + 2ma^2 = 4ma^2$$

By coordinate permutation symmetry, the remaining diagonal elements I_{yy} and I_{zz} have identical values:

$$I_{yy} = 4ma^2$$

$$I_{zz} = 4ma^2$$

Calculating the off-diagonal product of inertia element $I_{xy} = -\sum_\alpha m x_\alpha y_\alpha$ gives:

$$I_{xy} = -m\left[(a)(0) + (0)(a) + (0)(0) + (a)(a)\right]$$

Evaluating the sum of products inside the brackets:

$$I_{xy} = -m(0 + 0 + 0 + a^2) = -ma^2$$

By spatial symmetry, all remaining off-diagonal elements are equal:

$$I_{xz} = I_{zx} = -ma^2$$

$$I_{yz} = I_{zy} = -ma^2$$

$$I_{yx} = -ma^2$$

Assembling these components, the inertia tensor matrix $\mathbf{I}$ about the origin is:

$$\mathbf{I} = ma^2 \begin{pmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ -1 & -1 & 4 \end{pmatrix}$$

To find the principal moments of inertia, we determine the eigenvalues of $\mathbf{I}$ by solving the characteristic equation:

$$\det(\mathbf{I} - I_{principal}\mathbf{E}) = 0$$

Let $I_{principal} = \lambda ma^2$, where $\mathbf{E}$ is the 3×3 identity matrix. Dividing the determinant equation by ma^2 yields:

$$\det \begin{pmatrix} 4-\lambda & -1 & -1 \\ -1 & 4-\lambda & -1 \\ -1 & -1 & 4-\lambda \end{pmatrix} = 0$$

Defining $x = 4 - \lambda$, the secular determinant simplifies to:

$$\det \begin{pmatrix} x & -1 & -1 \\ -1 & x & -1 \\ -1 & -1 & x \end{pmatrix} = 0$$

Expanding the determinant along the first row:

$$x[x(x) - (-1)(-1)] - (-1)[(-1)(x) - (-1)(-1)] + (-1)[(-1)(-1) - x(-1)] = 0$$

Simplifying the algebraic expression yields:

$$x(x^2-1)+1(-x-1)-1(1+x)=0$$

Factoring out the term $(x+1)$:

$$x(x+1)(x-1)-2(x+1)=0$$

$$(x+1)[x(x-1)-2]=0$$

Expanding the terms inside the brackets:

$$(x+1)(x^2-x-2)=0$$

Factoring the quadratic polynomial $(x^2-x-2)=(x-2)(x+1)$ gives:

$$(x+1)^2(x-2)=0$$

The roots of the characteristic equation are:

$$x_1=2, \quad x_2=-1, \quad x_3=-1$$

Using $\lambda=4-x$, the eigenvalue solutions for λ are:

$$\lambda_1=4-2=2$$

$$\lambda_2=4-(-1)=5$$

$$\lambda_3=4-(-1)=5$$

Multiplying by ma^2, the principal moments of inertia are obtained as:

$$I_1=2ma^2$$

$$I_2=5ma^2$$

$$I_3=5ma^2$$

To determine the principal axis associated with the unique principal moment $I_1=2ma^2$ ($\lambda_1=2$), we solve the matrix eigenvalue equation $(\mathbf{I}-2ma^2\mathbf{E})\mathbf{v}=0$ for vector $\mathbf{v}=(u_1,u_2,u_3)^T$:

$$ma^2\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix}\begin{pmatrix}u_1\\u_2\\u_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

Writing out the first two algebraic equations:

$$2u_1-u_2-u_3=0$$

$$-u_1+2u_2-u_3=0$$

Subtracting the second equation from the first equation gives:

$$3u_1-3u_2=0 \implies u_1=u_2$$

Substituting $u_1=u_2$ back into the first equation yields:

$$2u_1-u_1-u_3=0 \implies u_3=u_1$$

Thus, the eigenvector corresponding to $I_1=2ma^2$ points along the symmetry axis $(1,1,1)$.

For the doubly degenerate principal moment $I_2 = I_3 = 5ma^2$ ($\lambda_2 = \lambda_3 = 5$), we set up the eigenvalue equation $(\mathbf{I} - 5ma^2\mathbf{E})\mathbf{v} = 0$:

$$ma^2 \begin{pmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This matrix equation reduces to a single linear constraint on the eigenvector components:

$$u_1 + u_2 + u_3 = 0$$

This constraint defines a plane perpendicular to the vector $(1, 1, 1)$. Consequently, any two orthogonal directions lying within this plane perpendicular to the $(1, 1, 1)$ axis serve as principal axes corresponding to the doubly degenerate principal moment of inertia $5ma^2$.

3. Euler's Angles

Definition of Euler Angles and Transformation Matrix

To specify the orientation of a rigid body in three-dimensional space relative to a fixed coordinate system, three independent angular parameters are required. These are known as the Euler angles, denoted by ϕ, θ, and ψ.

Let $Oxyz$ be a fixed space coordinate system with origin O, and let $Ox'y'z'$ be a coordinate system fixed in the moving rigid body with the same origin O. The orientation of the body frame $Ox'y'z'$ relative to the space frame $Oxyz$ is achieved through three successive right-handed rotations about specific axes.

The first rotation is performed about the space z-axis through an angle ϕ (known as the precession angle, where $0 \leq \phi \leq 2\pi$). This transforms the space frame $Oxyz$ into an intermediate frame $Ox_1y_1z_1$. The z_1-axis coincides with the original z-axis. The new x_1-axis formed by this rotation is called the line of nodes, which marks the intersection of the space xy-plane and the body $x'y'$-plane. The transformation matrix $\mathbf{D}$ for this rotation is:

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

The second rotation is performed about the line of nodes (x_1-axis) through an angle θ (known as the nutation angle, where $0 \leq \theta \leq \pi$). This transforms the frame $Ox_1y_1z_1$ into a second intermediate frame $Ox_2y_2z_2$. The x_2-axis coincides with the line of nodes x_1, while the z_2-axis is rotated into alignment with the final body axis z'. The transformation matrix $\mathbf{C}$ for this rotation is:

$$\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

The third rotation is performed about the body z'-axis (z_2-axis) through an angle ψ (known as the body spin angle, where $0 \leq \psi \leq 2\pi$). This transforms the second intermediate frame $Ox_2y_2z_2$ into the final body-fixed frame $Ox'y'z'$ (also referred to as axes 1, 2, and 3). The transformation matrix $\mathbf{B}$ for this rotation is:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

The total transformation taking space coordinates (x, y, z) to body coordinates (x', y', z') is given by the matrix product $\mathbf{A} = \mathbf{BCD}$.

Multiplying the matrices $\mathbf{C}$ and $\mathbf{D}$ first gives:

$$\mathbf{CD} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Performing the matrix multiplication yields:

$$\mathbf{CD} = \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\cos\theta\sin\phi & \cos\theta\cos\phi & \sin\theta \\ \sin\theta\sin\phi & -\sin\theta\cos\phi & \cos\theta \end{pmatrix}$$

Multiplying matrix $\mathbf{B}$ by the result $\mathbf{CD}$ gives the full transformation matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\cos\theta\sin\phi & \cos\theta\cos\phi & \sin\theta \\ \sin\theta\sin\phi & -\sin\theta\cos\phi & \cos\theta \end{pmatrix}$$

Evaluating all matrix elements results in the orthogonal matrix of direction cosines $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} \cos\psi\cos\phi - \cos\theta\sin\phi\sin\psi & \cos\psi\sin\phi + \cos\theta\cos\phi\sin\psi & \sin\theta\sin\psi \\ -\sin\psi\cos\phi - \cos\theta\sin\phi\cos\psi & -\sin\psi\sin\phi + \cos\theta\cos\phi\cos\psi & \sin\theta\cos\psi \\ \sin\theta\sin\phi & -\sin\theta\cos\phi & \cos\theta \end{pmatrix}$$

The elements A_{ij} of this matrix are the direction cosines relating the body axes (x', y', z') to the space axes (x, y, z).

Components of Angular Velocity Along the Body Axes

The total angular velocity vector $\boldsymbol{\omega}$ of the rigid body is the vector sum of three angular velocity vectors corresponding to the time derivatives of the three Euler angles: $\dot{\boldsymbol{\phi}}$, $\dot{\boldsymbol{\theta}}$, and $\dot{\boldsymbol{\psi}}$.

The angular velocity vector $\dot{\boldsymbol{\phi}}$ is directed along the space z-axis. The angular velocity vector $\dot{\boldsymbol{\theta}}$ is directed along the line of nodes (x_1-axis). The angular velocity vector $\dot{\boldsymbol{\psi}}$ is directed along the body z'-axis (axis 3).

To find the components of the total angular velocity along the body axes $(1, 2, 3)$, each rate of rotation is resolved along the body axes separately.

First, the angular velocity $\dot{\boldsymbol{\psi}}$ acts directly along the body 3-axis (z'-axis), so its components along the body axes are:

$$\dot{\psi}_1 = 0$$

$$\dot{\psi}_2 = 0$$

$$\dot{\psi}_3 = \dot{\psi}$$

Second, the angular velocity $\dot{\boldsymbol{\theta}}$ acts along the line of nodes x_1 (which is identical to x_2). The body axes 1 and 2 lie in a plane rotated by angle ψ relative to the line of nodes about the 3-axis. Projecting $\dot{\boldsymbol{\theta}}$ onto the body 1 and 2 axes yields:

$$\dot{\theta}_1 = \dot{\theta}\cos\psi$$

$$\dot{\theta}_2 = -\dot{\theta}\sin\psi$$

$$\dot{\theta}_3 = 0$$

Third, the angular velocity $\dot{\boldsymbol{\phi}}$ acts along the space z-axis. Resolving $\dot{\boldsymbol{\phi}}$ into components along the second intermediate frame axes (x_2, y_2, z_2), where the angle between z and z_2 is θ, gives:

$$\dot{\phi}_{x_2} = 0$$

$$\dot{\phi}_{y_2} = \dot{\phi}\sin\theta$$

$$\dot{\phi}_{z_2} = \dot{\phi}\cos\theta$$

Rotating these intermediate components through the final angle ψ about the z_2-axis to reach the body axes $(1, 2, 3)$ gives:

$$\dot{\phi}_1 = \dot{\phi}_{x_2}\cos\psi + \dot{\phi}_{y_2}\sin\psi = 0 \cdot \cos\psi + (\dot{\phi}\sin\theta)\sin\psi = \dot{\phi}\sin\theta\sin\psi$$

$$\dot{\phi}_2 = -\dot{\phi}_{x_2}\sin\psi + \dot{\phi}_{y_2}\cos\psi = -0 \cdot \sin\psi + (\dot{\phi}\sin\theta)\cos\psi = \dot{\phi}\sin\theta\cos\psi$$

$$\dot{\phi}_3 = \dot{\phi}_{z_2} = \dot{\phi}\cos\theta$$

Summing the individual contributions along each body axis gives the total components of angular velocity $\omega_1, \omega_2, \omega_3$ along the principal body axes.

Adding the components along the body 1-axis gives:

$$\omega_1 = \dot{\phi}_1 + \dot{\theta}_1 + \dot{\psi}_1 = \dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi$$

Adding the components along the body 2-axis gives:

$$\omega_2 = \dot{\phi}_2 + \dot{\theta}_2 + \dot{\psi}_2 = \dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi$$

Adding the components along the body 3-axis gives:

$$\omega_3 = \dot{\phi}_3 + \dot{\theta}_3 + \dot{\psi}_3 = \dot{\phi}\cos\theta + \dot{\psi}$$

4. Euler's Equations of Motion

Derivation for a Rigid Body Rotating About a Fixed Point

Consider a rigid body rotating about a fixed point O. Let $Oxyz$ be an inertial space-fixed frame, and let $O123$ be a rotating frame fixed in the rigid body whose coordinate axes coincide with the principal axes of inertia passing through O. The principal moments of inertia associated with these axes are I_1, I_2, and I_3.

In the space-fixed frame, Newton's second law for rotational motion states that the time rate of change of the angular momentum vector $\mathbf{L}$ is equal to the applied external torque $\mathbf{N}$:

$$\left(\frac{d\mathbf{L}}{dt}\right)_{\text{space}} = \mathbf{N}$$

The rate of change of any vector $\mathbf{A}$ in the space frame is related to its rate of change in a frame rotating with angular velocity $\boldsymbol{\omega}$ by the operator relation:

$$\left(\frac{d\mathbf{A}}{dt}\right)_{\text{space}} = \left(\frac{d\mathbf{A}}{dt}\right)_{\text{body}} + \boldsymbol{\omega} \times \mathbf{A}$$

Applying this relation directly to the angular momentum vector $\mathbf{L}$ yields:

$$\left(\frac{d\mathbf{L}}{dt}\right)_{\text{body}} + \boldsymbol{\omega} \times \mathbf{L} = \mathbf{N}$$

In the body-fixed principal axis frame, the angular velocity $\boldsymbol{\omega}$, angular momentum $\mathbf{L}$, and external torque $\mathbf{N}$ are expanded in terms of the unit vectors $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3)$ along the principal axes:

$$\boldsymbol{\omega} = \omega_1\hat{\mathbf{e}}_1 + \omega_2\hat{\mathbf{e}}_2 + \omega_3\hat{\mathbf{e}}_3$$

$$\mathbf{L} = L_1\hat{\mathbf{e}}_1 + L_2\hat{\mathbf{e}}_2 + L_3\hat{\mathbf{e}}_3$$

$$\mathbf{N} = N_1\hat{\mathbf{e}}_1 + N_2\hat{\mathbf{e}}_2 + N_3\hat{\mathbf{e}}_3$$

Because the coordinate axes are the principal axes of inertia, the inertia tensor is diagonal, relating the components of $\mathbf{L}$ directly to $\boldsymbol{\omega}$:

$$L_1 = I_1\omega_1$$

$$L_2 = I_2\omega_2$$

$$L_3 = I_3\omega_3$$

The derivative of $\mathbf{L}$ with respect to time evaluated inside the body frame, where the unit vectors $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$ are stationary, is:

$$\left(\frac{d\mathbf{L}}{dt}\right)_{\text{body}} = \dot{L}_1\hat{\mathbf{e}}_1 + \dot{L}_2\hat{\mathbf{e}}_2 + \dot{L}_3\hat{\mathbf{e}}_3 = I_1\dot{\omega}_1\hat{\mathbf{e}}_1 + I_2\dot{\omega}_2\hat{\mathbf{e}}_2 + I_3\dot{\omega}_3\hat{\mathbf{e}}_3$$

The cross product $\boldsymbol{\omega} \times \mathbf{L}$ evaluated in component form is:

$$\boldsymbol{\omega} \times \mathbf{L} = (\omega_2 L_3 - \omega_3 L_2)\hat{\mathbf{e}}_1 + (\omega_3 L_1 - \omega_1 L_3)\hat{\mathbf{e}}_2 + (\omega_1 L_2 - \omega_2 L_1)\hat{\mathbf{e}}_3$$

Substituting the relations $L_1 = I_1\omega_1$, $L_2 = I_2\omega_2$, and $L_3 = I_3\omega_3$ into the cross product gives:

$$\boldsymbol{\omega} \times \mathbf{L} = (\omega_2 I_3\omega_3 - \omega_3 I_2\omega_2)\hat{\mathbf{e}}_1 + (\omega_3 I_1\omega_1 - \omega_1 I_3\omega_3)\hat{\mathbf{e}}_2 + (\omega_1 I_2\omega_2 - \omega_2 I_1\omega_1)\hat{\mathbf{e}}_3$$

Factoring out the angular velocity components in each term yields:

$$\boldsymbol{\omega} \times \mathbf{L} = (I_3 - I_2)\omega_2\omega_3\hat{\mathbf{e}}_1 + (I_1 - I_3)\omega_3\omega_1\hat{\mathbf{e}}_2 + (I_2 - I_1)\omega_1\omega_2\hat{\mathbf{e}}_3$$

Substituting both $(d\mathbf{L}/dt)_{\text{body}}$ and $\boldsymbol{\omega} \times \mathbf{L}$ back into the vector equation of motion gives:

$$[I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3]\hat{\mathbf{e}}_1 + [I_2\dot{\omega}_2 + (I_1 - I_3)\omega_3\omega_1]\hat{\mathbf{e}}_2 + [I_3\dot{\omega}_3 + (I_2 - I_1)\omega_1\omega_2]\hat{\mathbf{e}}_3 = N_1\hat{\mathbf{e}}_1 + N_2\hat{\mathbf{e}}_2 + N_3\hat{\mathbf{e}}_3$$

Equating components along each of the three principal axes produces Euler's equations of motion for a rigid body rotating about a fixed point:

For the 1-axis:

$$I_1\dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = N_1$$

For the 2-axis:

$$I_2\dot{\omega}_2 - (I_3 - I_1)\omega_3\omega_1 = N_2$$

For the 3-axis:

$$I_3\dot{\omega}_3 - (I_1 - I_2)\omega_1\omega_2 = N_3$$

Force-Free Motion

For force-free motion, the external net torque acting on the body is zero:

$$\mathbf{N} = 0$$

This requires all individual torque components along the body principal axes to vanish:

$$N_1 = 0$$

$$N_2 = 0$$

$$N_3 = 0$$

Setting the right-hand sides of Euler's equations of motion to zero gives the system of equations describing the force-free motion of a rigid body:

For the 1-axis:

$$I_1\dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = 0$$

For the 2-axis:

$$I_2\dot{\omega}_2 - (I_3 - I_1)\omega_3\omega_1 = 0$$

For the 3-axis:

$$I_3\dot{\omega}_3 - (I_1 - I_2)\omega_1\omega_2 = 0$$

5. Coriolis Force

Rotating Frames of Reference and Operator Relation

Consider an inertial reference frame S and a non-inertial reference frame S' whose origin coincides with that of S, but which rotates with an instantaneous angular velocity $\boldsymbol{\omega}$ relative to S. To formulate the laws of dynamics in the rotating frame S', it is necessary to relate time derivatives of vectors evaluated in the inertial frame to those evaluated in the rotating frame.

Let $\mathbf{A}(t)$ be an arbitrary vector field. During an infinitesimal time interval dt, the vector changes due to two separate effects: the change of its components within the rotating frame, and the physical rotation of the reference coordinate axes themselves. The change of vector $\mathbf{A}$ as observed in the inertial space, $(d\mathbf{A})_{\text{in}}$, is related to the change observed within the rotating frame, $(d\mathbf{A})_{\text{rot}}$, by the vector cross product describing spatial rotation:

$$(d\mathbf{A})_{\text{in}} = (d\mathbf{A})_{\text{rot}} + (\boldsymbol{\omega} \times \mathbf{A})dt$$

Dividing this differential relation by the scalar time increment dt yields the operational relation connecting time rates of change in the two frames:

$$\left(\frac{d\mathbf{A}}{dt}\right)_{\text{in}} = \left(\frac{d\mathbf{A}}{dt}\right)_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{A}$$

This operator identity applies to any vector quantity. Expressed in operator form, the relation is:

$$\left(\frac{d}{dt}\right)_{\text{in}} = \left(\frac{d}{dt}\right)_{\text{rot}} + \boldsymbol{\omega}\times$$

Derivation of the Equation of Motion in a Rotating Frame

Let $\mathbf{r}(t)$ be the position vector of a particle of mass m. Applying the operator transformation rule to the position vector $\mathbf{r}$ yields the velocity in the inertial frame:

$$\left(\frac{d\mathbf{r}}{dt}\right)_{\text{in}} = \left(\frac{d\mathbf{r}}{dt}\right)_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r}$$

Defining the velocity observed in the inertial frame as $\mathbf{v}_{\text{in}} = (d\mathbf{r}/dt)_{\text{in}}$ and the velocity observed in the rotating frame as $\mathbf{v}_{\text{rot}} = (d\mathbf{r}/dt)_{\text{rot}}$, the relation becomes:

$$\mathbf{v}_{\text{in}} = \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r}$$

To determine the acceleration in the inertial frame, $\mathbf{a}_{\text{in}} = (d\mathbf{v}_{\text{in}}/dt)_{\text{in}}$, we apply the operator identity to the inertial velocity vector $\mathbf{v}_{\text{in}}$:

$$\mathbf{a}_{\text{in}} = \left(\frac{d\mathbf{v}_{\text{in}}}{dt}\right)_{\text{in}} = \left(\frac{d\mathbf{v}_{\text{in}}}{dt}\right)_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{v}_{\text{in}}$$

Substituting the expression $\mathbf{v}_{\text{in}} = \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r}$ into the first term on the right-hand side yields:

$$\left(\frac{d\mathbf{v}_{\text{in}}}{dt}\right)_{\text{rot}} = \left(\frac{d}{dt}\right)_{\text{rot}} (\mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r})$$

Expanding the time derivative of the sum in the rotating frame using the product rule gives:

$$\left(\frac{d\mathbf{v}_{\text{in}}}{dt}\right)_{\text{rot}} = \left(\frac{d\mathbf{v}_{\text{rot}}}{dt}\right)_{\text{rot}} + \left(\frac{d\boldsymbol{\omega}}{dt}\right)_{\text{rot}} \times \mathbf{r} + \boldsymbol{\omega} \times \left(\frac{d\mathbf{r}}{dt}\right)_{\text{rot}}$$

Substituting $(d\mathbf{r}/dt)_{\text{rot}} = \mathbf{v}_{\text{rot}}$ into the final term simplifies this expression to:

$$\left(\frac{d\mathbf{v}_{\text{in}}}{dt}\right)_{\text{rot}} = \left(\frac{d\mathbf{v}_{\text{rot}}}{dt}\right)_{\text{rot}} + \left(\frac{d\boldsymbol{\omega}}{dt}\right)_{\text{rot}} \times \mathbf{r} + \boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}$$

Next, substituting the expression for $\mathbf{v}_{\text{in}}$ into the cross-product term $\boldsymbol{\omega} \times \mathbf{v}_{\text{in}}$ yields:

$$\boldsymbol{\omega} \times \mathbf{v}_{\text{in}} = \boldsymbol{\omega} \times (\mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{r})$$

Distributing the cross product over the addition gives:

$$\boldsymbol{\omega} \times \mathbf{v}_{\text{in}} = \boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

Combining the terms for $(d\mathbf{v}_{\text{in}}/dt)_{\text{rot}}$ and $\boldsymbol{\omega} \times \mathbf{v}_{\text{in}}$ produces the total acceleration in the inertial frame:

$$\mathbf{a}_{\text{in}} = \left(\frac{d\mathbf{v}_{\text{rot}}}{dt}\right)_{\text{rot}} + \left(\frac{d\boldsymbol{\omega}}{dt}\right)_{\text{rot}} \times \mathbf{r} + \boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

Combining the two identical terms $\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}$ gives:

$$\mathbf{a}_{\text{in}} = \left(\frac{d\mathbf{v}_{\text{rot}}}{dt}\right)_{\text{rot}} + 2\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + \left(\frac{d\boldsymbol{\omega}}{dt}\right)_{\text{rot}} \times \mathbf{r}$$

Newton's second law of motion holds in the inertial frame S, stating that the real applied force $\mathbf{F}$ equals mass times acceleration:

$$\mathbf{F} = m\mathbf{a}_{\text{in}}$$

Substituting the expanded expression for $\mathbf{a}_{\text{in}}$ into Newton's second law gives:

$$\mathbf{F} = m\left(\frac{d\mathbf{v}_{\text{rot}}}{dt}\right)_{\text{rot}} + 2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} + m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + m\left(\frac{d\boldsymbol{\omega}}{dt}\right)_{\text{rot}} \times \mathbf{r}$$

Rearranging this relation to isolate the effective rate of change of momentum observed in the rotating frame, $m(d\mathbf{v}_{\text{rot}}/dt)_{\text{rot}}$, yields the complete equation of motion in the rotating reference frame:

$$m\left(\frac{d\mathbf{v}_{\text{rot}}}{dt}\right)_{\text{rot}} = \mathbf{F} - 2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) - m\left(\frac{d\boldsymbol{\omega}}{dt}\right)_{\text{rot}} \times \mathbf{r}$$

Identification of Coriolis and Centrifugal Forces

The equation of motion in the rotating frame resembles Newton's second law, provided that auxiliary vector terms are treated as apparent or fictitious forces acting on the particle:

1. **Coriolis Force:** The velocity-dependent inertial force given by:

$$\mathbf{F}_{\text{Cor}} = -2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}$$

1. **Centrifugal Force:** The position-dependent inertial force directed radially outward from the axis of rotation given by:

$$\mathbf{F}_{\text{cent}} = -m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

1. **Euler Force:** The inertial force arising from non-uniform rotation:

$$\mathbf{F}_{\text{Euler}} = -m \left(\frac{d\boldsymbol{\omega}}{dt} \right)_{\text{rot}} \times \mathbf{r}$$

When the angular velocity $\boldsymbol{\omega}$ of the rotating frame is uniform in time, $(d\boldsymbol{\omega}/dt)_{\text{rot}} = 0$, and the equation of motion reduces to:

$$m \left(\frac{d\mathbf{v}_{\text{rot}}}{dt} \right)_{\text{rot}} = \mathbf{F} - 2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

Physical Interpretation and Effect on Earth

The Coriolis force $-2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}$ arises entirely because the velocity of a particle is being measured relative to a rotating frame of reference. Because of the cross product, the Coriolis force exhibits the following fundamental properties: - It acts perpendicular to both the instantaneous velocity vector $\mathbf{v}_{\text{rot}}$ and the angular velocity vector $\boldsymbol{\omega}$. - Because $\mathbf{F}_{\text{Cor}} \cdot \mathbf{v}_{\text{rot}} = 0$, the Coriolis force does no mechanical work on the moving particle; it changes only the direction of motion, not the speed. - It is proportional to the speed v_{rot} of the particle relative to the rotating frame, vanishing completely for a particle at rest in the rotating frame ($\mathbf{v}_{\text{rot}} = 0$).

Particle Moving on the Rotating Earth

Consider a particle moving near the surface of the Earth, which rotates about its polar axis from west to east with angular speed $\omega \approx 7.292 \times 10^{-5}\ \text{rad/s}$. Let λ denote the latitude of a point on the Earth's surface (positive in the Northern Hemisphere).

At latitude λ, the Earth's angular velocity vector $\boldsymbol{\omega}$ can be resolved into local reference frame components. Let the local coordinate axes be $\hat{\mathbf{i}}$ pointing East, $\hat{\mathbf{j}}$ pointing North, and $\hat{\mathbf{k}}$ pointing vertically upward (zenith). In this local coordinate system, the angular velocity vector of the Earth is:

$$\boldsymbol{\omega} = 0\hat{\mathbf{i}} + \omega\cos\lambda\hat{\mathbf{j}} + \omega\sin\lambda\hat{\mathbf{k}}$$

Suppose a particle moves horizontally with velocity $\mathbf{v}_{\text{rot}} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}}$. The Coriolis acceleration vector is:

$$\mathbf{a}_{\text{Cor}} = -2\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}$$

Evaluating the vector cross product explicitly:

$$\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & \omega\cos\lambda & \omega\sin\lambda \\ v_x & v_y & 0 \end{vmatrix}$$

Expanding the determinant gives:

$$\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}} = (-\omega v_y \sin\lambda)\hat{\mathbf{i}} + (\omega v_x \sin\lambda)\hat{\mathbf{j}} - (\omega v_x \cos\lambda)\hat{\mathbf{k}}$$

Multiplying by -2 yields the Coriolis acceleration components:

$$\mathbf{a}_{\text{Cor}} = (2\omega v_y \sin\lambda)\hat{\mathbf{i}} - (2\omega v_x \sin\lambda)\hat{\mathbf{j}} + (2\omega v_x \cos\lambda)\hat{\mathbf{k}}$$

Analyzing the horizontal components of this acceleration: - If a particle moves North ($v_y > 0, v_x = 0$), the horizontal Coriolis acceleration is $+2\omega v_y \sin\lambda\hat{\mathbf{i}}$, causing a deflection to the East (to the right of the direction of motion in the Northern

Hemisphere). - If a particle moves East ($v_x > 0, v_y = 0$), the horizontal Coriolis acceleration is $-2\omega v_x \sin\lambda \hat{\mathbf{j}}$, causing a deflection to the South (again to the right in the Northern Hemisphere).

Thus, in the Northern Hemisphere ($\sin\lambda > 0$), any moving body is deflected to its right. In the Southern Hemisphere ($\sin\lambda < 0$), the sign of $\sin\lambda$ reverses, causing deflection to the left of the path of motion. This phenomenon governs large-scale atmospheric circulation, producing counter-clockwise cyclonic winds around low-pressure systems in the Northern Hemisphere and clockwise circulation in the Southern Hemisphere.

6. Force-Free Motion of a Symmetrical Top

Euler's Equations for a Torque-Free Symmetric Top

A rigid body is defined as a symmetrical top if two of its principal moments of inertia relative to the center of mass (or a fixed point) are equal. Let the principal axes fixed in the body be denoted by x_1, x_2, x_3, with corresponding principal moments of inertia I_1, I_2, I_3. For a symmetrical top, we have:

$$I_1 = I_2 \neq I_3$$

The x_3-axis is designated as the principal axis of symmetry of the top.

Euler's dynamic equations governing the components of angular velocity $(\omega_1, \omega_2, \omega_3)$ along the principal axes fixed in the body are:

$$I_1 \frac{d\omega_1}{dt} - (I_2 - I_3)\omega_2\omega_3 = N_1$$

$$I_2 \frac{d\omega_2}{dt} - (I_3 - I_1)\omega_3\omega_1 = N_2$$

$$I_3 \frac{d\omega_3}{dt} - (I_1 - I_2)\omega_1\omega_2 = N_3$$

where N_1, N_2, N_3 are the components of external torque along the principal body axes.

For force-free and torque-free motion, no external forces or torques act on the top:

$$N_1 = 0, \quad N_2 = 0, \quad N_3 = 0$$

Substituting $N_3 = 0$ and $I_1 = I_2$ into the third Euler equation gives:

$$I_3 \frac{d\omega_3}{dt} - (I_1 - I_1)\omega_1\omega_2 = 0$$

Since $(I_1 - I_1) = 0$, this simplifies directly to:

$$I_3 \frac{d\omega_3}{dt} = 0$$

Dividing by the non-zero moment of inertia I_3 yields:

$$\frac{d\omega_3}{dt} = 0$$

Integrating with respect to time shows that the component of angular velocity along the symmetry axis remains constant throughout the motion:

$$\omega_3 = \text{constant}$$

Precession Frequency in the Body Frame

Now consider the remaining two Euler equations. Substituting $N_1 = 0$ and $I_2 = I_1$ into the first equation:

$$I_1 \frac{d\omega_1}{dt} - (I_1 - I_3)\omega_2\omega_3 = 0$$

Isolating the term $I_1 \frac{d\omega_1}{dt}$ gives:

$$I_1 \frac{d\omega_1}{dt} = (I_1 - I_3)\omega_2\omega_3$$

Dividing both sides by I_1 yields:

$$\frac{d\omega_1}{dt} = \frac{I_1 - I_3}{I_1}\omega_3\omega_2$$

Factoring out a negative sign in the numerator gives:

$$\frac{d\omega_1}{dt} = -\left(\frac{I_3 - I_1}{I_1}\omega_3\right)\omega_2$$

Similarly, substituting $N_2 = 0$ into the second Euler equation gives:

$$I_1 \frac{d\omega_2}{dt} - (I_3 - I_1)\omega_3\omega_1 = 0$$

Isolating the term $I_1 \frac{d\omega_2}{dt}$ gives:

$$I_1 \frac{d\omega_2}{dt} = (I_3 - I_1)\omega_3\omega_1$$

Dividing both sides by I_1 yields:

$$\frac{d\omega_2}{dt} = \left(\frac{I_3 - I_1}{I_1}\omega_3\right)\omega_1$$

We define the constant angular frequency Ω of torque-free precession in the body frame as:

$$\Omega = \frac{I_3 - I_1}{I_1}\omega_3$$

Using this definition, the coupled differential equations for the transverse components ω_1 and ω_2 simplify to:

$$\frac{d\omega_1}{dt} = -\Omega\omega_2$$

$$\frac{d\omega_2}{dt} = \Omega\omega_1$$

Solution for the Angular Velocity Components

To decouple these equations, differentiate the first differential equation with respect to time t:

$$\frac{d^2\omega_1}{dt^2} = -\Omega\frac{d\omega_2}{dt}$$

Substituting the expression for $d\omega_2/dt = \Omega\omega_1$ into this derivative gives:

$$\frac{d^2\omega_1}{dt^2} = -\Omega(\Omega\omega_1)$$

Simplifying the right-hand side yields:

$$\frac{d^2\omega_1}{dt^2} = -\Omega^2\omega_1$$

Rearranging into standard simple harmonic oscillator form gives:

$$\frac{d^2\omega_1}{dt^2} + \Omega^2\omega_1 = 0$$

The general solution for $\omega_1(t)$ is:

$$\omega_1(t) = \omega_0 \cos(\Omega t + \alpha)$$

where ω_0 represents the amplitude of the transverse angular velocity and α is an arbitrary phase constant determined by initial conditions.

To find $\omega_2(t)$, isolate ω_2 from the relation $d\omega_1/dt = -\Omega\omega_2$:

$$\omega_2(t) = -\frac{1}{\Omega}\frac{d\omega_1}{dt}$$

Differentiating $\omega_1(t)$ with respect to time gives:

$$\frac{d\omega_1}{dt} = -\Omega\omega_0 \sin(\Omega t + \alpha)$$

Substituting this derivative into the expression for $\omega_2(t)$ yields:

$$\omega_2(t) = -\frac{1}{\Omega}\left(-\Omega\omega_0 \sin(\Omega t + \alpha)\right)$$

Simplifying the product gives:

$$\omega_2(t) = \omega_0 \sin(\Omega t + \alpha)$$

The magnitude of the transverse angular velocity vector $\boldsymbol{\omega}_\perp = \omega_1\hat{\mathbf{e}}_1 + \omega_2\hat{\mathbf{e}}_2$ is evaluated by taking the sum of squares:

$$\omega_\perp^2 = \omega_1^2(t) + \omega_2^2(t)$$

Substituting the solutions for $\omega_1(t)$ and $\omega_2(t)$:

$$\omega_\perp^2 = \omega_0^2\cos^2(\Omega t + \alpha) + \omega_0^2\sin^2(\Omega t + \alpha)$$

Factoring out ω_0^2 gives:

$$\omega_\perp^2 = \omega_0^2\left(\cos^2(\Omega t + \alpha) + \sin^2(\Omega t + \alpha)\right)$$

Using the trigonometric identity $\cos^2\theta + \sin^2\theta = 1$:

$$\omega_\perp^2 = \omega_0^2$$

Taking the positive square root confirms that the transverse angular speed is constant:

$$\omega_\perp = \omega_0 = \text{constant}$$

The total magnitude of the angular velocity vector $\boldsymbol{\omega}$ is:

$$\omega = \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2}$$

Substituting $\omega_1^2 + \omega_2^2 = \omega_0^2$ yields:

$$\omega = \sqrt{\omega_0^2 + \omega_3^2} = \text{constant}$$

Body Cone, Space Cone, and Motion of the Symmetry Axis

The Body Cone

In the body-fixed frame of reference, the component ω_3 along the symmetry axis x_3 is constant, and the perpendicular vector component $\boldsymbol{\omega}_\perp$ rotates around the x_3-axis with constant angular frequency $\Omega = \frac{I_3 - I_1}{I_1}\omega_3$. Consequently, as seen from within the rotating rigid body, the angular velocity vector $\boldsymbol{\omega}$ traces out a circular cone centered on the symmetry axis x_3. This locus is called the **body cone**. The semi-vertical angle γ_b of the body cone is given by:

$$\tan\gamma_\text{b} = \frac{\omega_\perp}{\omega_3} = \frac{\omega_0}{\omega_3}$$

The Space Cone and Conservation Laws

In an inertial (space-fixed) frame, because the net external torque is zero ($\mathbf{N} = 0$), the total angular momentum vector $\mathbf{L}$ is strictly conserved in both magnitude and direction:

$$\mathbf{L} = \text{constant vector in space}$$

Furthermore, the rotational kinetic energy T is conserved:

$$T = \frac{1}{2}\boldsymbol{\omega}\cdot\mathbf{L} = \text{constant}$$

Expanding the dot product $\boldsymbol{\omega}\cdot\mathbf{L} = \omega L\cos\gamma_\text{s}$, where γ_s is the angle between the instantaneous angular velocity $\boldsymbol{\omega}$ and the fixed angular momentum vector $\mathbf{L}$:

$$\cos\gamma_\text{s} = \frac{2T}{\omega L} = \text{constant}$$

Because ω, L, and T are all constant in time, the angle γ_s between $\boldsymbol{\omega}$ and $\mathbf{L}$ remains fixed. Therefore, in the space frame, the angular velocity vector $\boldsymbol{\omega}$ precesses around the fixed directional axis of $\mathbf{L}$, tracing out a second cone known as the **space cone**.

Geometric Motion: Rolling Cones

The instantaneous motion of the symmetrical top can be described geometrically as the **body cone rolling without slipping on the fixed space cone**: - The line of contact between the two cones is the instantaneous axis of rotation, directed along the vector $\boldsymbol{\omega}$. - If $I_3 > I_1$ (an oblate top, such as a disk), the precession frequency $\Omega = \frac{I_3 - I_1}{I_1}\omega_3 > 0$ has the same sign as ω_3, meaning the body cone lies outside the space cone (direct precession). - If $I_3 < I_1$ (a prolate top, such as a rod), $\Omega < 0$, meaning the body cone lies inside the space cone (retrograde precession).

Motion of the Symmetry Axis

The symmetry axis x_3 of the top, the instantaneous angular velocity $\boldsymbol{\omega}$, and the constant angular momentum vector $\mathbf{L}$ always lie in a single plane. As $\boldsymbol{\omega}$ precesses, the symmetry axis x_3 sweeps out a circular path around the fixed direction of $\mathbf{L}$ in space at a constant rate known as the space precession frequency Ω_space, given by:

$$\Omega_\text{space} = \frac{L}{I_1}$$

Thus, in the absence of external torques, the symmetry axis of a rotating symmetrical top executes a steady uniform precession about the space-fixed angular momentum vector.

7. Small Oscillations: Equilibrium and Potential Energy

Consider a conservative mechanical system described by n generalized coordinates $q_1, q_2, \ldots, q_n$. Suppose the system possesses a position of stable equilibrium, which without loss of generality can be chosen as the origin of the coordinate system, such that $q_i = 0$ for all $i = 1, 2, \ldots, n$.

The potential energy $V(q_1, q_2, \ldots, q_n)$ of the system can be expanded in a Taylor series about the equilibrium configuration $q_i = 0$:

$$V(q_1, q_2, \ldots, q_n) = V(0, \ldots, 0) + \sum_{i=1}^{n} \left.\frac{\partial V}{\partial q_i}\right|_0 q_i + \frac{1}{2} \sum_{i=1}^{n}\sum_{j=1}^{n} \left.\frac{\partial^2 V}{\partial q_i \partial q_j}\right|_0 q_i q_j + \ldots$$

The potential energy at the equilibrium position is a constant baseline value, which can be set to zero without affecting the equations of motion:

$$V_0 = V(0, \ldots, 0) = 0$$

At a position of equilibrium, the generalized forces acting on the system must vanish, which means the first partial derivatives of the potential energy with respect to each coordinate evaluated at equilibrium are identically zero:

$$\left.\frac{\partial V}{\partial q_i}\right|_0 = 0 \quad (i = 1, 2, \ldots, n)$$

We define the potential energy coefficients V_{ij} as the second partial derivatives evaluated at the equilibrium configuration:

$$V_{ij} = \left.\frac{\partial^2 V}{\partial q_i \partial q_j}\right|_0$$

Because the order of partial differentiation is interchangeable for smooth functions, these potential energy coefficients form a symmetric matrix:

$$V_{ij} = V_{ji}$$

Truncating the Taylor series expansion to second order in small displacement quantities q_i, the potential energy of the system takes the quadratic form:

$$V = \frac{1}{2} \sum_{i=1}^{n}\sum_{j=1}^{n} V_{ij} q_i q_j$$

For a system with stationary constraints, the kinetic energy T is generally a quadratic form in the generalized velocities $\dot{q}_i$:

$$T = \frac{1}{2} \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}(q_1, \ldots, q_n) \dot{q}_i \dot{q}_j$$

Expanding the coefficient functions $M_{ij}(q)$ in a Taylor series about the equilibrium position $q_k = 0$ yields:

$$M_{ij}(q_1, \ldots, q_n) = M_{ij}(0, \ldots, 0) + \sum_{k=1}^{n} \left.\frac{\partial M_{ij}}{\partial q_k}\right|_0 q_k + \ldots$$

Since the generalized velocities $\dot{q}_i$ are already small quantities of first order, retaining terms in kinetic energy up to second order overall requires approximating $M_{ij}(q)$ by its constant value evaluated at equilibrium:

$$T_{ij} = M_{ij}(0, \ldots, 0)$$

Because of the symmetry in the kinetic energy summation over velocity indices, these constant coefficients satisfy:

$$T_{ij} = T_{ji}$$

Substituting these constant coefficients into the kinetic energy expression yields the second-order kinetic energy function:

$$T = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} T_{ij} \dot{q}_i \dot{q}_j$$

The equations of motion governing small oscillations about the stable equilibrium configuration are obtained directly from these quadratic kinetic and potential energy forms, giving for each coordinate index i:

$$\sum_{j=1}^{n} (T_{ij} \ddot{q}_j + V_{ij} q_j) = 0 \quad (i = 1, 2, \ldots, n)$$

8. Frequencies of Free Vibration and Normal Coordinates

To find the natural modes of oscillation for the system of coupled linear differential equations, we look for periodic trial solutions where all generalized coordinates oscillate with the same angular frequency ω and a common phase angle ϕ:

$$q_j(t) = A_j \cos(\omega t + \phi)$$

Differentiating this trial solution twice with respect to time yields the generalized acceleration for coordinate q_j:

$$\ddot{q}_j(t) = -\omega^2 A_j \cos(\omega t + \phi)$$

Substituting $q_j(t)$ and $\ddot{q}_j(t)$ into the system's equations of motion gives:

$$\sum_{j=1}^{n} (-\omega^2 T_{ij} A_j + V_{ij} A_j) \cos(\omega t + \phi) = 0$$

Since this relationship must hold at all times t, the time-dependent factor $\cos(\omega t + \phi)$ can be canceled, leaving a set of homogeneous algebraic linear equations for the unknown amplitude coefficients A_j:

$$\sum_{j=1}^{n} (V_{ij} - \omega^2 T_{ij}) A_j = 0 \quad (i = 1, 2, \ldots, n)$$

For non-trivial solutions (where not all amplitudes A_j are zero), the determinant of the coefficient matrix must vanish, yielding the secular equation:

$$\det(V - \omega^2 T) = 0$$

Expanding this determinant gives an n-th degree algebraic equation in ω^2, known as the characteristic equation:

$$\begin{vmatrix} V_{11} - \omega^2 T_{11} & V_{12} - \omega^2 T_{12} & \cdots & V_{1n} - \omega^2 T_{1n} \\ V_{21} - \omega^2 T_{21} & V_{22} - \omega^2 T_{22} & \cdots & V_{2n} - \omega^2 T_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ V_{n1} - \omega^2 T_{n1} & V_{n2} - \omega^2 T_{n2} & \cdots & V_{nn} - \omega^2 T_{nn} \end{vmatrix} = 0$$

Solving the secular equation yields n roots for ω^2, denoted by $\omega_1^2, \omega_2^2, \ldots, \omega_n^2$. The positive square roots ω_k represent the characteristic frequencies or eigenfrequencies of free vibration of the system.

Each root ω_k corresponds to a normal mode of vibration. Substituting a specific eigenfrequency ω_k back into the algebraic system determines the ratio of amplitudes A_{jk} for that mode:

$$\sum_{j=1}^{n}(V_{ij} - \omega_k^2 T_{ij})A_{jk} = 0 \quad (i = 1, 2, \ldots, n)$$

The most general motion of the system is a superposition of all n normal modes:

$$q_j(t) = \sum_{k=1}^{n} C_k A_{jk} \cos(\omega_k t + \phi_k)$$

To decouple the system of equations, we transform from the original generalized coordinates q_j to a new set of independent coordinates Q_k, called normal coordinates. The physical coordinates q_j are related to the normal coordinates Q_k via a linear transformation:

$$q_j = \sum_{k=1}^{n} a_{jk} Q_k$$

Under this transformation, the kinetic energy matrix T_{ij} and potential energy matrix V_{ij} are simultaneously diagonalized. Expressed in terms of normal coordinates, the kinetic energy simplifies to:

$$T = \frac{1}{2}\sum_{k=1}^{n} \dot{Q}_k^2$$

Similarly, the potential energy expressed in normal coordinates reduces to a sum of independent square terms:

$$V = \frac{1}{2}\sum_{k=1}^{n} \omega_k^2 Q_k^2$$

The equation of motion for each individual normal coordinate Q_k is completely decoupled from all other normal coordinates:

$$\ddot{Q}_k + \omega_k^2 Q_k = 0 \quad (k = 1, 2, \ldots, n)$$

Thus, each normal coordinate Q_k oscillates independently with a single, unique natural frequency ω_k:

$$Q_k(t) = B_k \cos(\omega_k t + \phi_k)$$

9. Longitudinal Vibration of a Linear Triatomic Molecule

Consider a linear symmetric triatomic molecule consisting of two end atoms each of mass m and a central atom of mass M aligned along a straight line. The atoms are connected by two identical springs, each having a force constant k.

Let x_1, x_2, x_3 represent the small longitudinal displacements of mass m (left atom), mass M (central atom), and mass m (right atom) from their respective equilibrium positions along the line of symmetry.

The total kinetic energy T of the molecule is the sum of the translational kinetic energies of the three individual masses:

$$T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}M\dot{x}_2^2 + \frac{1}{2}m\dot{x}_3^2$$

The potential energy V is the potential energy stored in the two stretched or compressed springs:

$$V = \frac{1}{2}k(x_2 - x_1)^2 + \frac{1}{2}k(x_3 - x_2)^2$$

Expanding the quadratic terms in the potential energy expression gives:

$$V = \frac{1}{2}k(x_1^2 - 2x_1x_2 + x_2^2) + \frac{1}{2}k(x_2^2 - 2x_2x_3 + x_3^2)$$

Collecting like terms converts the potential energy into matrix form:

$$V = \frac{1}{2}kx_1^2 + kx_2^2 + \frac{1}{2}kx_3^2 - kx_1x_2 - kx_2x_3$$

From the kinetic and potential energy expressions, the kinetic energy matrix T and potential energy matrix V are written as:

$$T = \begin{pmatrix} m & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & m \end{pmatrix}, \quad V = \begin{pmatrix} k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{pmatrix}$$

The secular equation $\det(V - \omega^2 T) = 0$ for determining the frequencies of free vibration takes the form:

$$\begin{vmatrix} k - m\omega^2 & -k & 0 \\ -k & 2k - M\omega^2 & -k \\ 0 & -k & k - m\omega^2 \end{vmatrix} = 0$$

Expanding this 3×3 determinant along its first row yields:

$$(k - m\omega^2)\left[(2k - M\omega^2)(k - m\omega^2) - (-k)(-k)\right] - (-k)\left[(-k)(k - m\omega^2) - 0\right] = 0$$

Simplifying the products within the brackets gives:

$$(k - m\omega^2)\left[(2k - M\omega^2)(k - m\omega^2) - k^2\right] - k^2(k - m\omega^2) = 0$$

Factoring out the common term $(k - m\omega^2)$ from the entire equation gives:

$$(k - m\omega^2)\left[(2k - M\omega^2)(k - m\omega^2) - 2k^2\right] = 0$$

Expanding the expression inside the square brackets gives:

$$(2k - M\omega^2)(k - m\omega^2) - 2k^2 = 2k^2 - 2km\omega^2 - Mk\omega^2 + Mm\omega^4 - 2k^2$$

Canceling the constant terms $2k^2$ leaves:

$$Mm\omega^4 - k(2m + M)\omega^2 = \omega^2\left[mM\omega^2 - k(2m + M)\right]$$

Substituting this back into the factored determinant equation yields the complete characteristic equation:

$$\omega^2(k - m\omega^2)\left[mM\omega^2 - k(2m + M)\right] = 0$$

Setting each factor to zero determines the three characteristic frequencies of free vibration:

First, setting the factor $\omega^2 = 0$ gives the first mode frequency:

$$\omega_1^2 = 0$$

This zero-frequency mode corresponds to a rigid body translation of the entire molecule along its axis, in which all three masses move together with no relative displacement and no spring deformation.

Second, setting the factor $(k - m\omega^2) = 0$ gives the second mode frequency:

$$\omega_2^2 = \frac{k}{m}$$

This mode corresponds to an antisymmetric stretch where the central mass remains completely at rest while the two outer masses oscillate out of phase with equal amplitudes.

Third, setting the factor $\left[mM\omega^2 - k(2m + M)\right] = 0$ gives the third mode frequency:

$$\omega_3^2 = k\left(\frac{1}{m} + \frac{2}{M}\right)$$

This mode corresponds to a symmetric stretch where the two outer masses move symmetrically in phase in one direction while the central mass moves in the opposite direction, preserving the center of mass of the molecule.

To construct the normal coordinates Q_1, Q_2, Q_3, we express the normal modes as orthogonal linear combinations of the physical displacement coordinates x_1, x_2, x_3:

For the translational mode $\omega_1^2 = 0$, the motion corresponds to the displacement of the center of mass, giving the first normal coordinate:

$$Q_1 = \frac{mx_1 + Mx_2 + mx_3}{2m + M}$$

For the mode $\omega_2^2 = \frac{k}{m}$, the central mass is stationary ($x_2 = 0$) and $x_1 = -x_3$, giving the second normal coordinate:

$$Q_2 = \frac{1}{\sqrt{2}}(x_1 - x_3)$$

For the symmetric mode $\omega_3^2 = k\left(\frac{1}{m} + \frac{2}{M}\right)$, the outer masses move together while the central mass moves oppositionally to keep the center of mass fixed, giving the third normal coordinate:

$$Q_3 = \frac{1}{\sqrt{2 + \frac{4m}{M}}}\left(x_1 - \frac{2m}{M}x_2 + x_3\right)$$

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No Assam University questions were found for this unit in the paper bank** — the questions below are from other universities' papers and standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] Define the inertia tensor.

Answer in brief: Section 2 — $I_{ij} = \sum m(\delta_{ij} r^2 - x_i x_j)$, a symmetric 3×3 matrix with $L_i = \sum_j I_{ij}\omega_j$.

S2. [standard] What are principal axes and principal moments of inertia?

Answer in brief: Section 2 — the axes in which the inertia tensor is diagonal; the diagonal entries I_1, I_2, I_3 are the principal moments, with $L_1 = I_1\omega_1$ and so on.

S3. [standard] Define Euler angles.

Answer in brief: Section 3 — ϕ precession about the space z-axis, θ nutation between the space and body z-axes, ψ spin about the body z-axis.

S4. [standard] State Euler's equations of motion for a rigid body.

Answer in brief: Section 4 — $I_1\dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = N_1$ and its cyclic permutations; for force-free motion the right-hand sides are zero.

S5. [standard] What is the Coriolis force?

Answer in brief: Section 5 — in a frame rotating with $\boldsymbol{\omega}$, a particle moving with velocity $\mathbf{v}$ feels $-2m\,\boldsymbol{\omega} \times \mathbf{v}$, deflecting it sideways to its motion.

S6. [standard] What are normal coordinates?

Answer in brief: Section 8 — linear combinations of the displacements in which each coordinate oscillates with a single normal-mode frequency, decoupling the small oscillations.

Long-form questions

L1. [non-AU] Derive Euler's equations for a rigid body rotating about a fixed point. What are Euler's equations of motion for the force-free motion of a rigid body? — CBCS-pattern UG Physics-I, Section B; St. Aloysius, M.Sc Sem-I, Nov 2021, Part A Q8.

What to write: Sections 1 and 4 in full, with Section 6 — from $(d\mathbf{L}/dt)_{space} = \mathbf{N}$ in the body frame, $I_1\dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = N_1$ (and cyclic); force-free means $N_1 = N_2 = N_3 = 0$.

L2. [non-AU] What are Euler angles? Describe the motion of a body in terms of direction cosines and Euler angles, and obtain the components of angular velocity along the body axes in terms of Euler angles. — St. Aloysius Nov 2021, Part C Q21; Christ College Nov 2025, Section C Q18.

What to write: Section 3 in full — the three angles and the three successive rotations, the direction cosines, and $\omega_1 = \dot{\phi}\sin\theta\sin\psi + \dot{\theta}\cos\psi$, $\omega_2 = \dot{\phi}\sin\theta\cos\psi - \dot{\theta}\sin\psi$, $\omega_3 = \dot{\phi}\cos\theta + \dot{\psi}$.

L3. [non-AU] Discuss the vibrations of a linear triatomic molecule. Give the expressions for kinetic energy and potential energy for its vibration and find the frequencies of free vibration and the normal coordinates. — Christ College Nov 2025, Section A Q5; St. Aloysius Nov 2021, Part A Q5.

What to write: Sections 7–9 in full — T and V for the three masses, the secular equation, and the three modes: $\omega^2 = 0$ (translation), $\omega^2 = k/m$ (antisymmetric stretch), $\omega^2 = k(1/m + 2/M)$ (symmetric stretch), with their normal coordinates.

L4. [non-AU] Four mass points each of mass m are placed at (a,0,0), (0,a,0), (0,0,a) and (a,a,a). Evaluate the inertia tensor of the system and find the principal axes and moments of inertia; also derive the equation of motion in a rotating frame and explain the Coriolis force. — St. Aloysius Nov 2021, Part B Q16; Gauhati Univ. B.Sc 1st Sem Physics 1.1, 2014, Q6a (rotating frame).

What to write: Sections 2 and 5 in full — the inertia tensor is ma^2 times the matrix with 4 on the diagonal and -1 off the diagonal; principal moments $2ma^2$ about (1,1,1) and $5ma^2$ (twice) about axes perpendicular to it; the rotating-frame equation with the Coriolis term $-2m\,\boldsymbol{\omega} \times \mathbf{v}$.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.